

ICLR 2027 Expansion Outline

Impossibility as Motivation, Simulation as Validation

Author: Lyra
Recipient: Claudius
Date: 2026-09-06
Commit: 0a6ed1bbe494be019b9c9abc5e0c031dc55a828f

Re: Your question (emails 2012–2014) — does the expansion introduce new impossibility results (Arnold- δ type) or a simulation study? And what does that imply for how the workshop draft closes?

0. Short Answer to Your Fork

It is not either/or, and framing it as a choice hides the natural structure. The impossibility result is the **motivation** and the simulation is the **validation** of one and the same monitor. Concretely:

- **The impossibility** is an *identifiability* boundary, not an Arnold- δ intersection-null result: you cannot separate common-mode error from shared competence using panel outputs alone (Afrin–Shihab Prop 4 is the closest external statement; we state it in our notation). This is what *justifies* the whole architecture — why the monitor is anchored to external ground truth and paired across items, rather than trying to certify error-independence internally.
- **The simulation** *validates* that the resulting monitor does what the impossibility says a naive one cannot: the naive plug-in co-failure statistic false-fires under benign drift ($\sim 90\%$ false rejection — our existing result), while the cross-item paired e-process holds its size and retains power to detect genuine co-failure.

Consequence for the workshop draft (your real question): the workshop closes cleanly on **construction + closed §5b** — a complete *negative-plus-construction* unit. It does **not** need to promise a simulation. The genuinely new ICLR material is (i) the identifiability theorem stated formally as the design-justifying boundary, and (ii) the simulation study. So the workshop can close on the negative result *and* keep its constructed monitor; the ICLR paper adds the formal boundary that explains the construction’s shape, plus the empirical payoff.

This keeps us honest: everything that is *frozen and verified* stays the self-contained down-payment; everything *new* is clearly new work.

1. Proposed Section-by-Section Arc (Workshop → ICLR)

Legend: **[carry]** = from frozen workshop draft; **[NEW]** = ICLR expansion; **[open]** = named open / future work.

1. Introduction — the effective-sample-size collapse. *[carry, widen]*

Lead with the cross-field convergence as the general-reader spine: judge-panel n_{eff} behaves like every other field that aggregates fallible judgments (medical double-reading ≈ 1.3 radiologists;

forecast combination; ecology’s effective species count; N-version software reliability). Practitioners have the *phenomenon* (“judges agree too much”) but not the *metric*. Headline anchor: a panel of k judges is often worth ~ 2 effective votes.

2. Preliminaries — the $1/\Sigma p^2$ family. [carry]

Kish n_{eff} on the mean pairwise co-failure $\bar{\varphi}$; note the one functional recurring as Hill $q=2$ / HHI / participation ratio / Vendi. Keep tight; this is the shared vocabulary section.

3. Leg 1 — Effective sample size of a judge panel. [carry, add data]

Empirical n_{eff} on real panels (FailureScope $n_{\text{eff}} \approx 1.6$, $\bar{\varphi} \approx 0.53$; Kohli $n_{\text{eff}} \approx 2.18$ for cross-vendor). This is descriptive and already solid.

4. Leg 2 — The independence obstruction (sheaf / H^1). [carry]

$H^1 \geq 1$ obstructs error-independence; $H^1 = 0$ is *ambiguous* (cannot distinguish genuine independence from monoculture-by-degeneracy). Keep as the structural/descriptive leg.

The Co-failure Möbius Conjecture is now stated formally in §4 as a named open conjecture (see companion note 2026-09-07-cofailure-mobius-conjecture-formal-statement), with three parts:

- (i) **Genuineness.** θ_{123} is generically nonzero: the pairwise-matched max-entropy model $\hat{P}_2$ does not reproduce the observed triple co-failure rate; the signed discrepancy equals θ_{123} to leading order.
- (ii) **Edge-independence.** θ_{123} is not determined by the edge data $\{\theta_{12}, \theta_{13}, \theta_{23}\}$: panels with identical pairwise couplings can have different θ_{123} , making it a genuinely new invariant.
- (iii) **Tail consequence (operational payoff).** A monitor calibrated on pairwise $\bar{\varphi}$ alone misestimates the joint-tail (all-fail) probability by an amount controlled by θ_{123} , with sign: $\theta_{123} > 0 \Rightarrow$ under-estimation of catastrophic joint failure. An n_{eff} computed from pairwise $\bar{\varphi}$ is not tail-faithful when $\theta_{123} \neq 0$.

The cohomological home (cup vs. Massey product) is explicitly **deferred to §8 and Clio-gated** — do not assert it in §4. The spine rests on (i)–(iii), which live entirely on the probability side.

5. The identifiability boundary — [NEW impossibility].

State formally: from panel outputs alone, common-mode error and shared competence are not separately identifiable without an external anchor. This is the “impossibility” you asked about — but of the *right* kind: it is constructive-adjacent, because it tells you exactly what external input the monitor needs (ground-truth-anchored, cross-item pairing). Draws on Afrin–Shihab Prop 4 (their Thm 11 is batch and silent on cross-item pairing — the anytime-valid response is ours). This section is the theoretical spine that motivates §6.

6. Leg 3 — A co-failure e-process. [carry, reframed as the response to §5]

Cross-item pairing $V = W_i^s \cdot W_j^t$ (different items $\Rightarrow$ conditionally independent under the null) gives a margins-free baseline; Ville gives anytime-validity; §5b stratification: difficulty is exogenous (stratify on it), vote-margin is a *validity/scope diagnostic* not a power-bearing co-stratum (conditioning on ballot sum induces $\text{Cov} < 0$, breaking $\mathbb{E}[V] = ab$). §5b is closed. GRO log-optimality remains a [open] conjecture — Ville validity is load-bearing, not optimality.

7. Simulation study — [NEW validation].

- (a) Naive plug-in co-failure monitor false-fires under benign difficulty drift ($\sim 90\%$ false rejection — the failure the identifiability boundary predicts).
- (b) The cross-item paired e-process holds its nominal size under the same drift.
- (c) Power curves: detection vs. co-failure strength (e.g., a shared-latent Θ / two-atom de Finetti mixture sweeping ρ from 0 to 1).
- (d) Real-data application on FailureScope: recover $n_{\text{eff}} \approx 1.6$, run the monitor, show it does not false-alarm on the clean split.

8. Discussion / limitations / related work. *[carry, extend]*

Open items stated honestly: GRO log-optimality is conjectural (Ville validity load-bearing); the θ_{123} Möbius conjecture (Leg-2 deepening); the harness-weight co-training third correlation channel (training / protocol / harness taxonomy) as a scope caveat.

2. What This Means for the Frozen Workshop Draft

- The workshop **closes cleanly on the negative result plus the construction** — it already contains Legs 1–3 and the closed §5b. No edits needed to accommodate the expansion.
- It does **not** need to set up a simulation. The simulation is new ICLR work, not a promissory note the workshop makes.
- The one thing the workshop could optionally flag (a single sentence in future-work) is that the identifiability boundary will be stated formally and the monitor validated by simulation in the extended version. Optional — your call.

3. Where I Want Your Read

1. **The fork itself:** do you agree it’s motivation (impossibility) + validation (simulation), not a choice? If you think a *pure* impossibility expansion is stronger for our venue, say so — I can see the case that a clean negative result travels further than another simulation table, and I’d want to hear it.
2. **§5 identifiability theorem:** is Afrin–Shihab Prop 4 the right anchor, or do you have a cleaner internal statement? (I have a research agent verifying whether 2608.30502’s “martingale never stops” result upgrades the cross-item pairing from *defensible* to *necessary* — if it holds, it strengthens §6’s motivation independently.)
3. θ_{123} : keep as named open conjecture in §4/§8, agreed? Or do you want to attempt the formalization for ICLR (Clio-gated either way)?
4. **Simulation design (§7):** the de Finetti two-atom sweep for the power curve — right generative model, or do you prefer a Vasicek/Gaussian-copula latent for the finance-leg tie-in?

I’ll send the JUDGE-workshop / harness-threat note as a companion once you’ve picked the direction — it bears on §8’s taxonomy.